

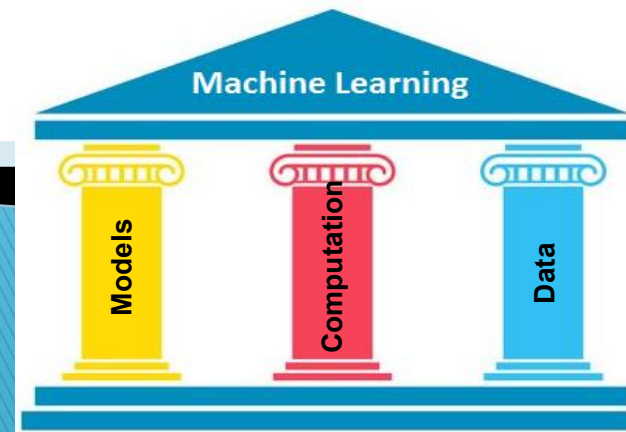
COMP1008 Part II: Machine Learning

- Machine learning I: [Intro](#)
- Machine learning II: Supervised vs. unsupervised learning
- Machine learning III: Supervised learning
- Machine learning IV: ANN & extensions

Please record your attendance today at [SEAtS](#)



Professor Rong Qu
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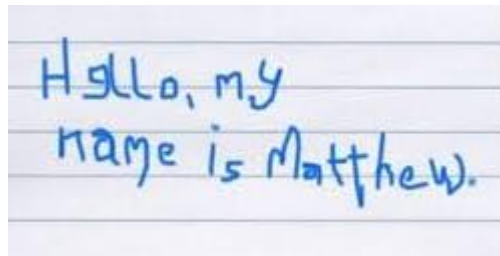


Machine Learning: Definition

A computer program is said to **learn** from **experience E** with respect to some class of **tasks T** and **performance measure P** if its performance at tasks in T, as measured by P, improves with experience E

Tom Mitchell

Let the **computer programs to learn** by themselves.



ID	Age	Sex	Job	Housing	SavingAcco	CheckAcco	CreditAmc	Duration	Purpose
995	31	female	1	own	little	NA	1736	12	furniture/equip
996	40	male	3	own	little	little	3857	30	car
10	25	female	2	rent	little	moderate	1295	12	car
17	25	male	2	own	NA	little	8072	30	business
999	27	male	2	own	moderate	moderate	4576	45	car

Machine Learning: To reach true AI

► Top-down / Classic (deductive)

1. Model all different functions and wire all these “agents” together
2. Write (manually) in programming language instructions to solve the (usually specific) task

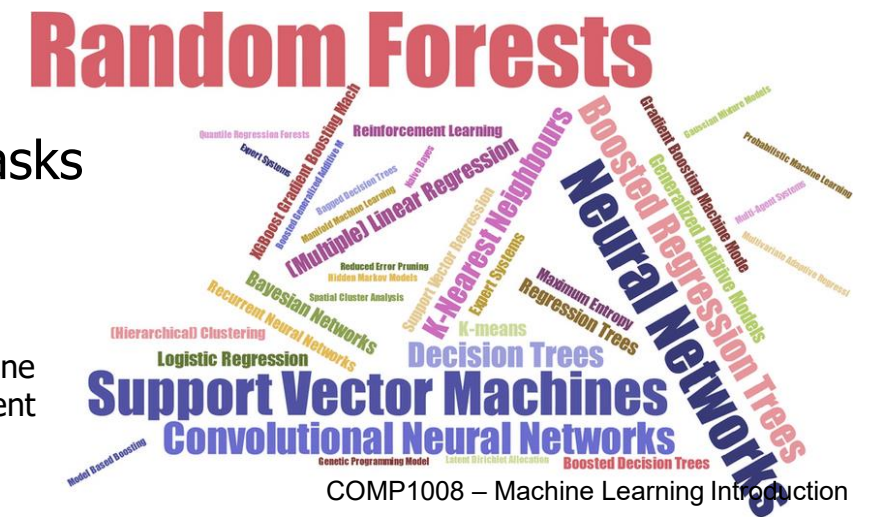
► Bottom-up / Machine learning (inductive)

- Give the system a lot of **data**, “so it can discover by itself concepts in the world”

Andrew Ng, Stanford University

1. Analyse and understand (manually) the task
2. Select the appropriate model(s)
3. Training and testing
4. Use the same model to (automatically) solve unseen tasks

MS Stupariu, S Cushman, AI Pleşoianu, C. Fürst, Machine learning in landscape ecological analysis: a review of recent approaches, Landscape Ecology 37(5), May 2022.



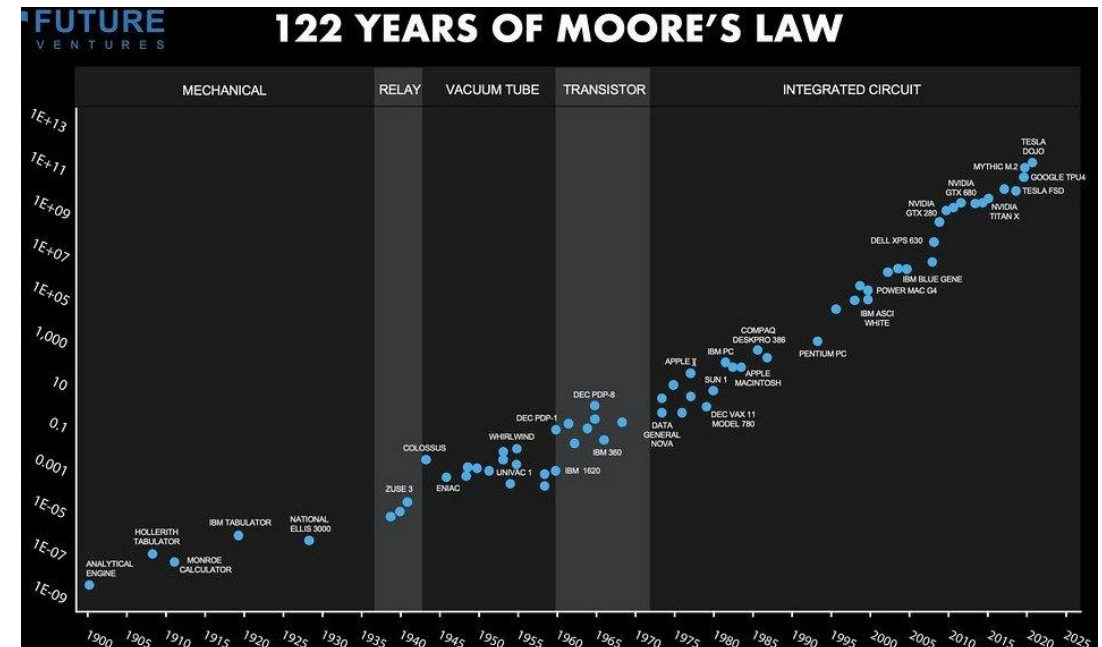
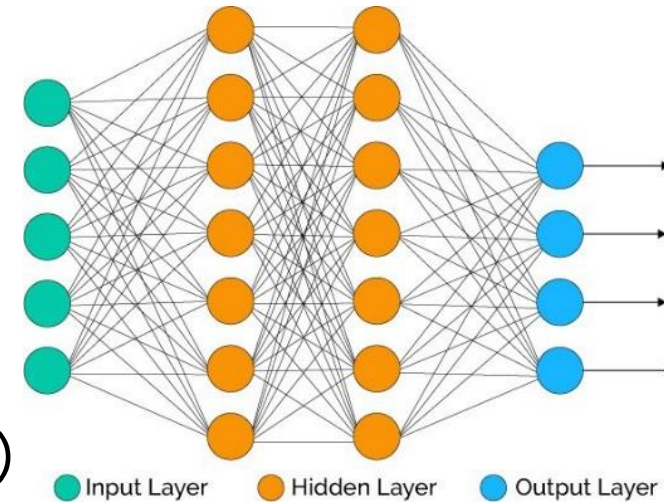
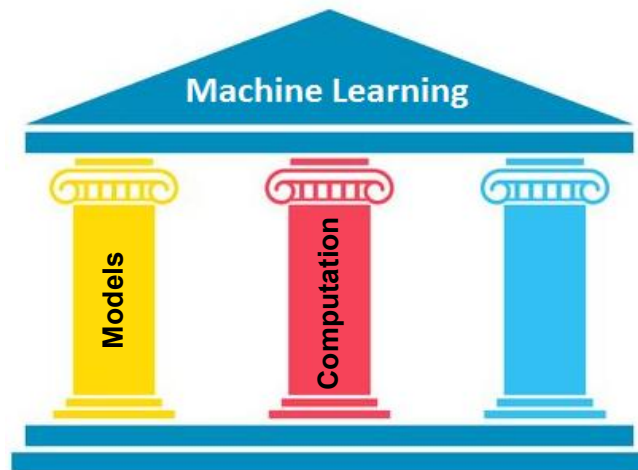
Machine Learning: Three pillars

1. Models and algorithms

- Doesn't require knowledge engineers to find all functions and wire them into an "agent" (i.e. **knowledge acquisition** bottleneck / modelling)

2. Powerful and cheaper computation

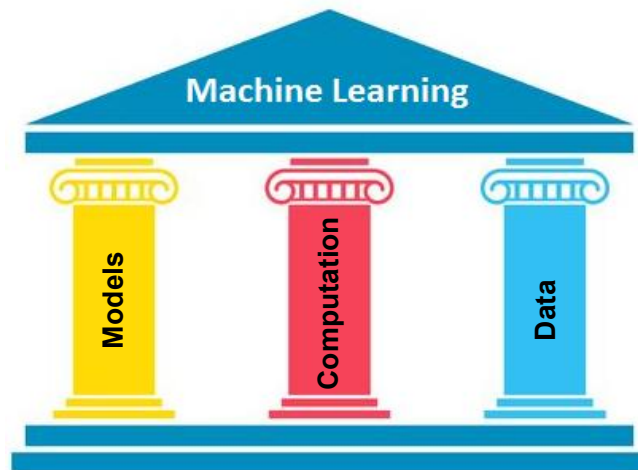
- Moore's Law(?)



Machine Learning: Three pillars

3. Massive data warehouse

- IoT / IoE: social media; business; industries; devices/vehicles; things ...
- Data in 2013 > all data in history
- Big data: quantity, format/dimension, speed



BIG DATA ANALYTICS



COMP1008

CS Course

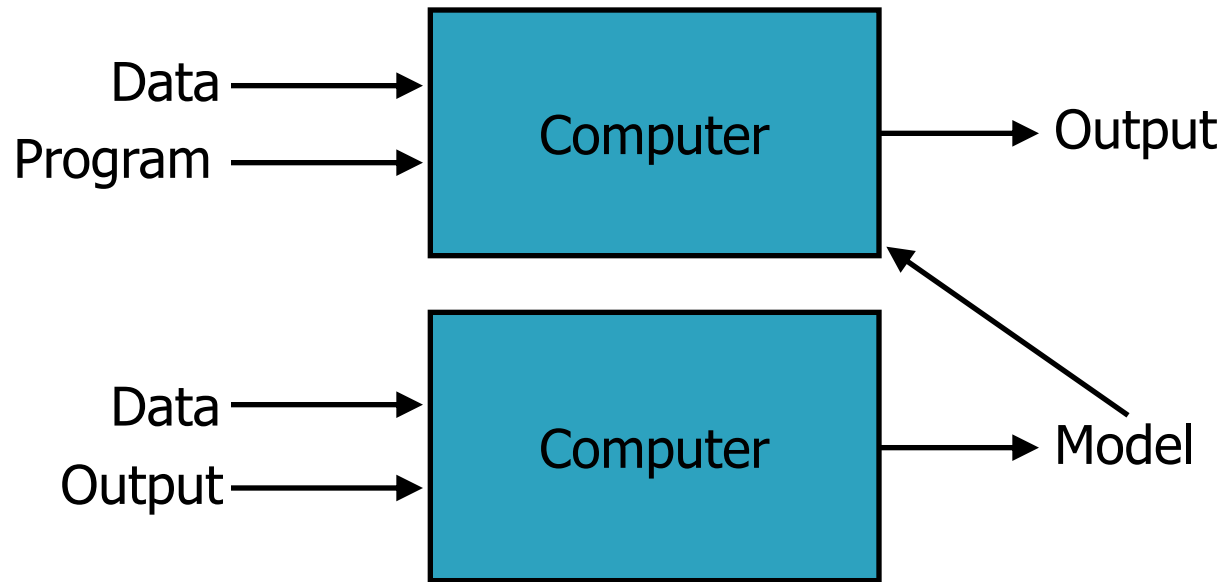
Machine learning

COMP4124 – Big Data Learning and Technologies



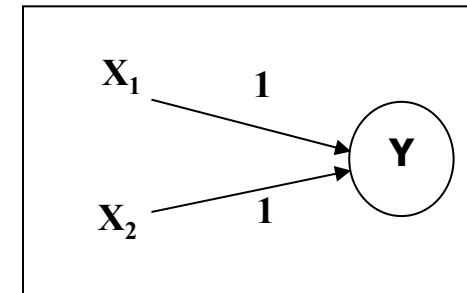
Google Books Ngram Viewer

Machine Learning vs. Traditional Programs



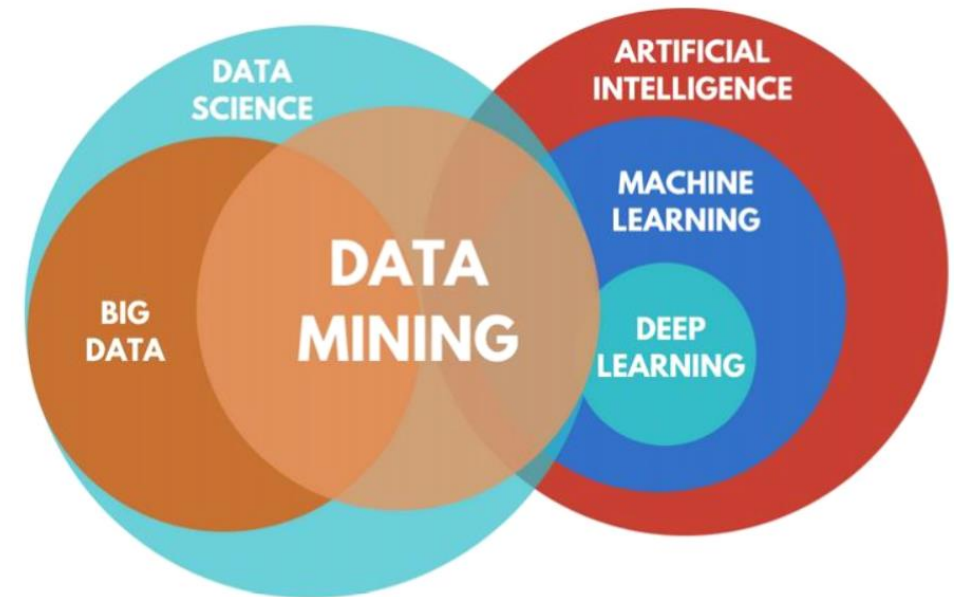
Tom Mitchell

AND		
X1	X2	Y
1	1	1
1	0	0
0	1	0
0	0	0

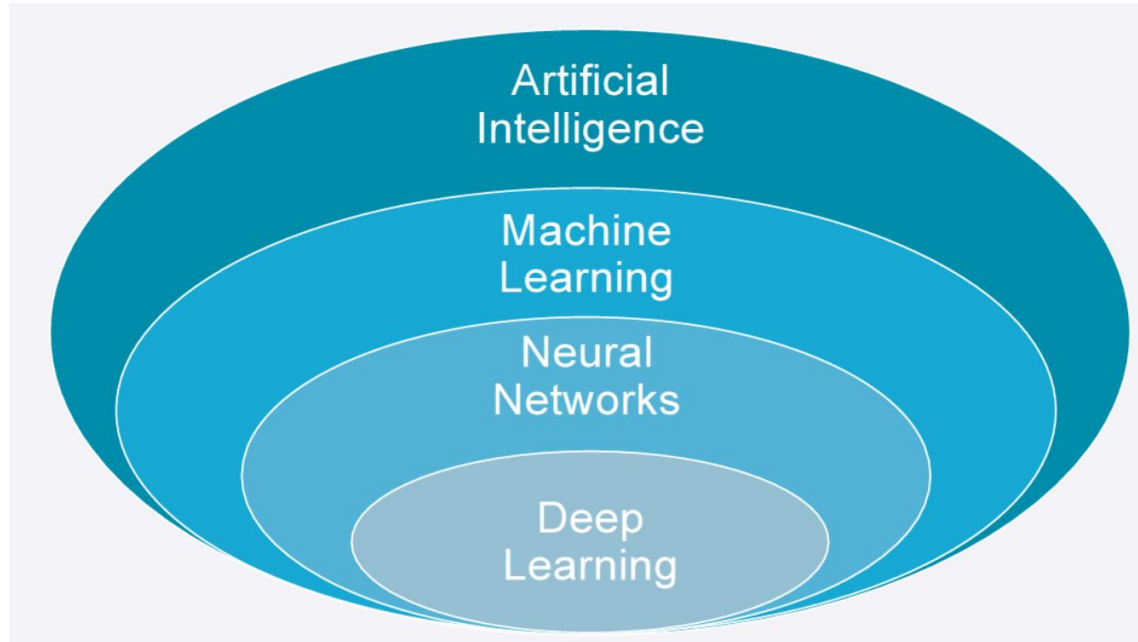


Machine Learning vs. Data Mining

- ▶ **Data mining:** exploration and analysis of large quantities of data to discover **valid**, **novel**, **useful**, and ultimately **understandable** patterns in data.
 - **Valid:** hold on new data with some certainty
 - **Novel:** non-obvious to the system
 - **Useful:** should be possible to act on the item
 - **Understandable:** humans can interpret the pattern/model
- ▶ Machine learning: predicts with **models**
- ▶ Data mining: explains **patterns**

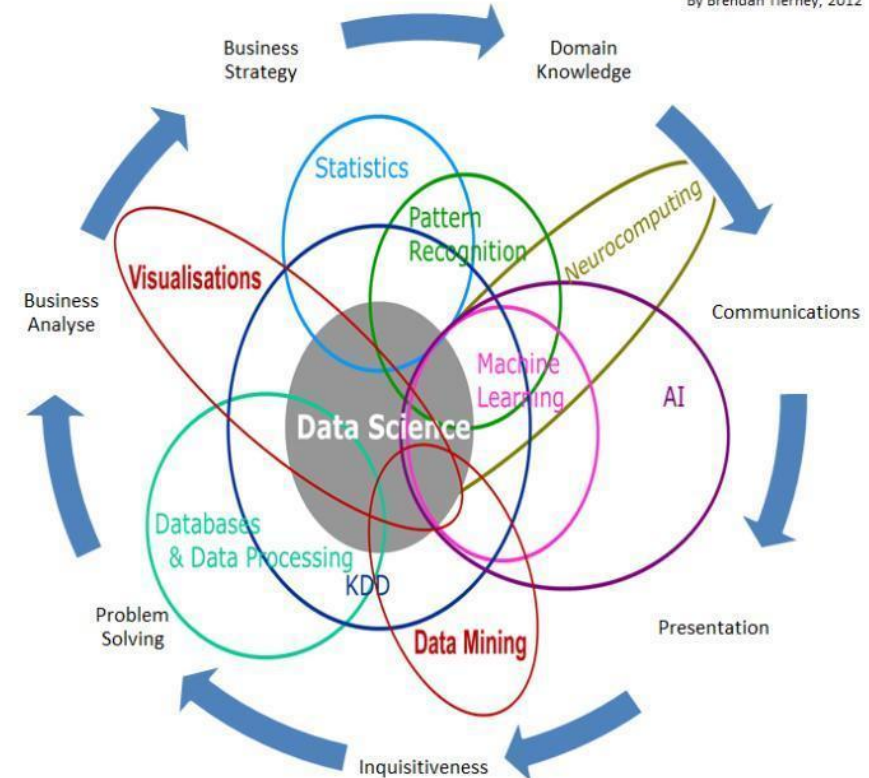


Machine Learning vs. Deep Learning vs. AI vs. Data Science



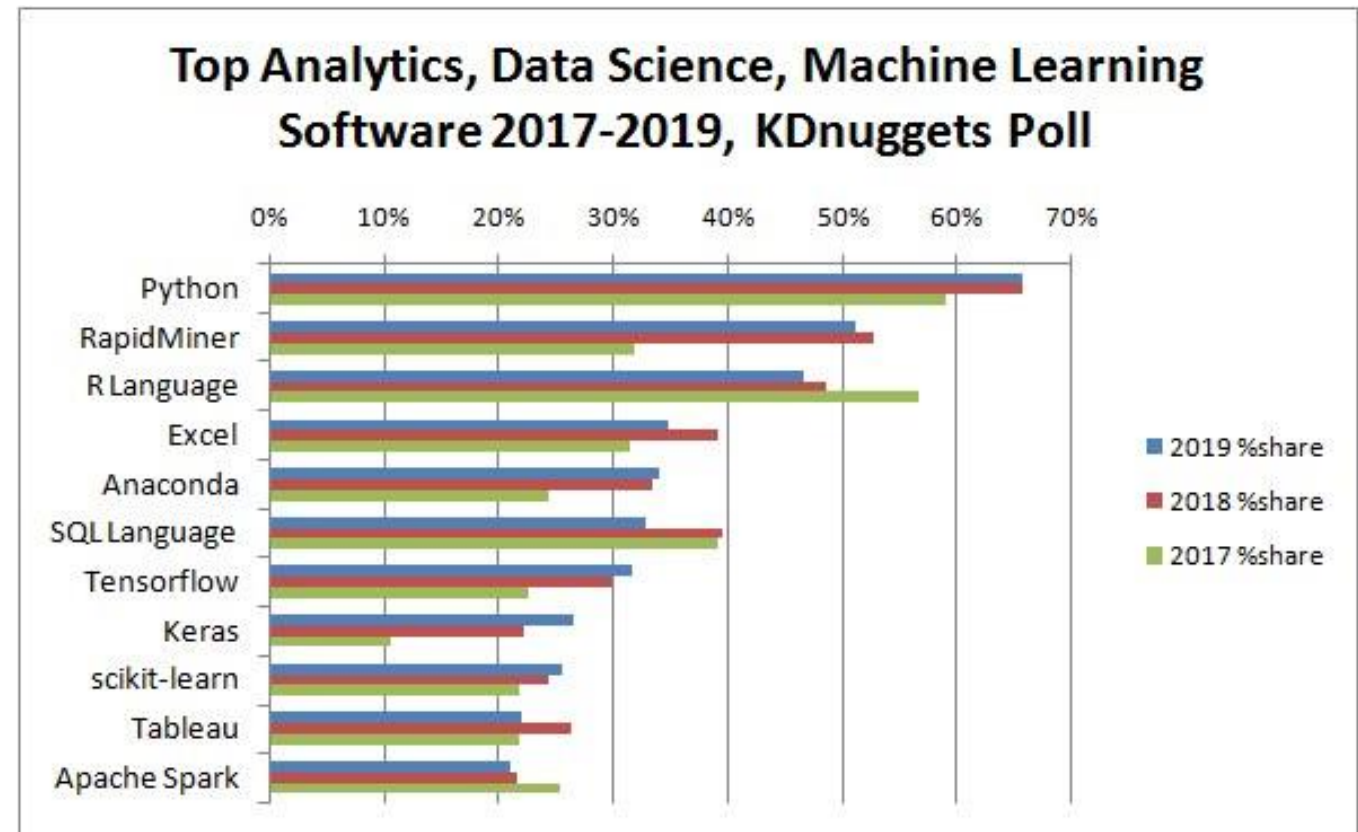
Data Science Is Multidisciplinary

By Brendan Tierney, 2012



Machine Learning: Datasets & Software

- ▶ [UCI Machine Learning repository](#)
- ▶ [KDnugget](#)
- ▶ [Kaggle](#) (also some free courses on data science)
- ▶ Python packages
 - Data **visualization**: Matplotlib, Seaborn
 - Data **manipulation**: NumPy, Pandas
 - **Machine learning**: Scikit-learn
 - Deep learning: TensorFlow, Keras, PyTorch
- ▶ Computing session 0



Next Week

- **Machine learning I:** Intro
 - **Machine learning II:** Un-supervised vs. Supervised learning
 - **Machine learning III:** Supervised learning
 - **Machine learning IV:** ANN & extension
 - Thursday: Computing session 0
 - Jupyter Notebook with Anaconda
 - Python basics
- 